Ryan Crane

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EDUCATION

Arizona State University, Barrett, the Honors College — Tempe, AZ
B.S.E. Mechanical Engineering, Minor in Data Science — **GPA 3.88** — Expected Fall 2027
M.S. Robotics and Autonomous Systems — Expected Fall 2028
Honors: Beckman Scholar (1 of 2 ASU undergraduates, 2025); iGEM Gold Medal (2024, 2025); National Merit Scholar

TECHNICAL SKILLS

CAD & Analysis: SolidWorks (50+ part assemblies), Ansys, FEniCS (FEA), MATLAB, LTspice, Cadence Virtuoso, KLayout
Prototyping & Mechatronics: 3D printing, lead-screw/stepper motion stages, CNC XY stages, robotic arm integration, fluidic plumbing, soldering and electronics assembly, equipment troubleshooting and repair
Software & Controls: Python, C++ (Arduino, ESP32), GRBL G-code, computer vision (segmentation models), OT-2 API
Imaging & Lab: Brightfield microscope design, SEM, confocal fluorescence microscopy, organoid/cell culture, PCR

EXPERIENCE

Laboratory Automation Engineer — Jan 2025 – Present
BioProtean Lab, Arizona State University — *Tempe, AZ*

- Sole engineer on a vision-guided automation system for organoid culture, owning it from concept through experimental testing: mechanical design, motion hardware, control software, and procurement within budget, working with faculty and staff scientists
- Designed and 3D-printed 50+ custom SolidWorks components for the microscope and automation-system housing
- Designed a motorized focusing stage (NEMA stepper + lead screw converting rotary to linear motion of the objective lens) and wrote the GRBL G-code autofocus routine
- Built a motorized well-scanning microscope with brightfield and widefield fluorescence modes; integrated a CNC XY stage that positions 96-well plates inside an Opentrons OT-2 pipetting robot
- Integrated an organoid segmentation model (Python) that locates targets and moves each well to a safe pipetting position; now integrating a Seeed reBot arm to shuttle plates to and from the incubator
- Automated nanobody production via phage-assisted continuous evolution with Python OT-2 protocols; repaired and restored lab instruments including a turbidostat and a surface plasmon resonance device

Electrical Engineering Research Assistant — May 2026 – Present
NSF Athena AI Institute for Edge Computing, Duke University — *Durham, NC*

- Designed a memristive CMOS synaptic circuit coupling two subthreshold silicon neurons (LIF and Izhikevich) using current mirrors, OTAs, and resistive non-volatile memory, minimizing area in Cadence Virtuoso; routed chip package in KLayout
- Tuned spiking behaviors with Monte Carlo simulation and built Python spiking-neural-network simulations comparing the design against simple and cascode mirror synapses under process variation and mismatch; co-authoring a conference submission

Synthetic Biology Hardware Team Lead (promoted from Hardware Engineer, Nov 2024) — Mar 2024 – Nov 2025
International Genetically Engineered Machine (iGEM) Team, Arizona State University — *Tempe, AZ*

- Built a continuous-culture bioreactor for phage-assisted continuous evolution (published as OptoPACE): designed 3D-printed housing and positive-pressure fluidic plumbing, soldered control circuits, and wrote Arduino C++ firmware
- Developed laser/photoresistor turbidity sensing and optimized data collection to generate mid-log growth curves
- Led weekly hardware meetings; designed a 3D-printed enclosure for a portable potentiostat-based electrochemical sensor and revised the design from stakeholder and expert interviews
- Gave oral talk and poster at the 2025 iGEM Grand Jamboree (Paris) to 100+ scientists and entrepreneurs; team won Gold

PROJECTS

Finite Element Modeling of Intestinal Organoid Mechanics (independent) — Jan – May 2026

- Replicated an intestinal organoid FEM in FEniCS; computed stress/strain fields under osmotic pressure and active contraction

AI-Assisted Defect Analysis for Ti-6Al-4V Aerospace Manufacturing (team) — Jan – May 2026

- Trained a segmentation model on Ti64 defect images and masks, and integrated it into an LLM-based fracture diagnosis interface

PUBLICATIONS & PRESENTATIONS

Crane, R. D.*, Isaacs, W. T.*, Ijaz, M.*, et al. (2026). *The OptoPACE Bioreactor: An Accessible Turbidostat System with Phage-Assisted Continuous Evolution Capability and Optogenetic Controls.*

9 conference presentations (2 oral), including *Machine Vision-Enabled Laboratory Automation for Scalable Organoid Culture*, 2026 Beckman Symposium, Irvine, CA